



EE 202P: Analog Electronics Lab

Course Code	EE 202P
Course Name	Analog Electronics Lab
L-T-P-C	0-0-2-1
Intended for	B.Tech.
Approval	5th Senate (Oct 2013)

Course Contents

- Lab 1: Introduction to Spice and Winspice
- Lab 2: Construct a spice netlist to study the characteristics of a BJT. Do a dc analysis and sweep the base current and illustrate the dependence of beta on collector current.
- Lab3: Construct spice netlist to analyse the sensitivity of Quiscent current for
 - a) Simple resistor biasing,
 - b) Voltage divider biasing,
 - c) Emitter degeneration biasing
 - d) Self biasing Prepare a report and discuss the biasing schemes based on your sensitivity analysis results.
- Lab 4: The common-emitter shown in the figure must amplify signals in the range of 1 MHz to 100 Mhz.
 - (a) Using the .op command, determine the bias conditions of Q1 and verify that it operates in the active region.
 - (b) Running an ac analysis, choose the value of C1 such that $|V_P/V_{in}| \sim 0.99$ at 1 MHz. This ensures that C1 acts as a short circuit at all frequencies of interest.
 - (c) Plot $|V_{out}/V_{in}|$ as a function of frequency for several values of C2 , e.g., 1 uF, 1 nF, and 1 pF. Determine the value of C2 such that the gain of the circuit at 10 MHz is only 2% below its maximum (i.e., for C2 = 1uF).
 - (d) With the proper value of C2 found in (c), determine the input impedance of the circuit at 10 MHz. (One approach is to insert a resistor in series with Vin and adjust its value until VP/Vin or Vout/Vin drops by a factor of two.)